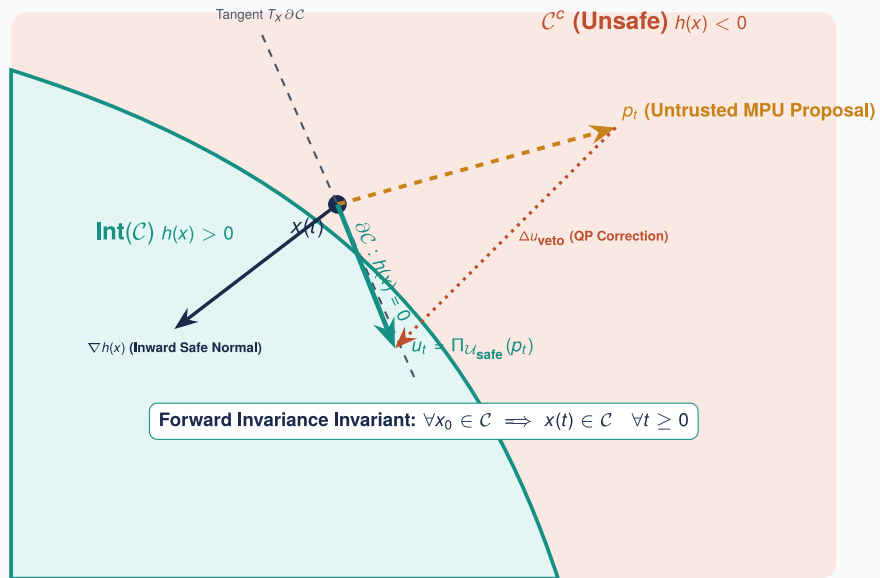


State Space: Forward Invariance $\mathcal{C} = \{x \in \mathbb{R}^n : h(x) \geq 0\}$

Nagumo's Theorem: Invariant trajectories satisfy $\dot{h}(x, u) \geq -\alpha(h(x))$ on boundary $\partial\mathcal{C}$.



Action Space: Minimal-Intervention Quadratic Program

Embedded Realization: Closed-form active-set KKT solver on Cortex-M4 in $< 100 \mu\text{s}$.

The Real-Time Safety Filter Formulation:

$$u_t^* = \arg \min_{u \in \mathcal{U}} \frac{1}{2} \|u - p_t\|_H^2$$

Subject to Lie Derivative Safety Invariant:

$$\underbrace{L_f h(x) + L_g h(x)u}_{\dot{h}(x, u)} \geq -\alpha(h(x))$$

Actuator Bounds: $u_{\min} \leq u \leq u_{\max}$ (Zero Heap, $\mathcal{O}(1)$ KKT Active-Set)

Forbidden (Breaches Invariant)

p_t (Unsafe Proposal)

$$\min \|u - p_t\|^2$$

$$u_t^* = \Pi_{\mathcal{U}_{\text{safe}}}(p_t)$$

$\mathcal{U}_{\text{safe}}(x)$ (Permitted Action Space)

Admissible Barrier Boundary: $L_g h(x)u = -L_f h(x) - \alpha(h(x))$

$$u = 0$$

MCU Execution Budget: Solve time $\leq 65 \mu\text{s}$ · Deterministic ISR · Zero Malloc

Three Inviolable Properties of the Minimal-Intervention CBF Filter:

1. Transparency: If $p_t \in \mathcal{U}_{\text{safe}}$, then $u_t = p_t$ (0% modification).

2. Minimality: If $p_t \notin \mathcal{U}_{\text{safe}}$, u_t is the least-squares

projection onto $\partial\mathcal{U}_{\text{safe}}$.

3. Forward Invariance: State $x(t)$ mathematically cannot cross $h(x) = 0$.